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RESEARCH ARTICLE

Substance Use and Educational Impacts in Youth With and Without Chronic Illness

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Introduction: Alcohol, cannabis, and nicotine are commonly used psychoactive substances that affect adolescent neurocognition. Little is known about the educational impacts of their use on measures of educational performance, participation and problems, especially among youth with a chronic illness who may use these substances to alleviate stress and symptoms.

Methods: Adolescents receiving general or subspecialty care were administered an electronic survey from 2016 to 2018. Data were analyzed in 2023. Using modified Poisson models, cross-sectional associations between past 12-month usage of alcohol, cannabis, and/or nicotine and educational impacts were estimated.

Results: Among 958 adolescents (mean age 16.0 years (SD 1.3), 564 (58.9%) female gender, 445 (46.5%) in subspecialty care), 294 (30.7%), 220 (23.0%), and 126 (13.2%) reported past 12-month use of alcohol, cannabis, and nicotine respectively, while 407 (42.5%) reported ≥1 educational impact, including recent lower grades 210 (21.9%), past 3-month truancy from school 164 (17.1%) or activities 170 (17.7%), and detention 82 (8.6%). Use of cannabis, but not other substances, was associated with negative educational impacts: lower grades (mostly C's/D's/F's), adjusted prevalence ratios [APR, (95% CI)] 1.54 (1.13−2.11); past 3-month truancy from school [2.16 (1.52−3.07)]; detention [2.29 (1.33−3.94)]. The association between cannabis use and any negative educational impact was stronger among adolescents with a chronic illness ($p<0.001$).

Conclusions: Among adolescents, cannabis use was associated with a heightened risk of negative educational impacts, even after controlling for alcohol and nicotine use. Adolescents with chronic illness were especially likely to experience negative educational impacts. Findings underscore need for preventive interventions and messaging to reduce risks.

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INTRODUCTION

Substance use is a global health concern that can hinder important developmental transitions during adolescence.[1] Alcohol, cannabis, and nicotine are the psychoactive substances most commonly used by adolescents in the U.S.,[2] and each is associated with decrements in functioning[3] and physiological impairment that can impact neurocognition.[4−9] For example, heavy episodic ("binge") drinking and heavy alcohol use are associated with poorer cognitive functioning on a broad range of neuropsychological assessments, including learning, psychomotor speed, attention, executive

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functioning, and impulsivity.[4,9] Cannabis may cause short-term memory impairment with heavy use; over time, adolescents who use cannabis heavily perform worse on tasks requiring flexible thinking and may experience reductions in motivation, attention, learning, memory, and executive functioning.[10−13] Nicotine is a stimulant that, in high doses, can impair attention and learning;[14] long-term exposure is associated with cognitive decline, lower psychomotor speed, and decreased cognitive flexibility.[7] Use of any of these substances is associated with increased rates of mental health problems, which can secondarily impact interest, effort and performance.[15]

Psychosocial impacts of substance use during adolescence are substantial, and decreased participation in important social, occupational, or recreational activities because of substance use is a general criterion for substance use disorder (i.e., regardless of substance).[16,17] Physiological and psychosocial impacts translate into harms. For example, alcohol use is strongly associated with accidents, injuries,[18] and risky sexual behavior.[19] Nicotine use, including vaping, is associated with respiratory problems and cannabis use has been associated with social and psychological problems, and with psychotic episodes.[20,21] Use of any of these substances has been associated with problems at school[17] and reduced educational attainment.[22]

Negative effects of substance use on adolescent neurocognitive and psychosocial functioning may be especially harmful for the one-of-four U.S. youth with a chronic medical condition (YCMC).[23] When using substances, YCMC are more likely to skip or miss prescribed medications,[24−27] and risks for poor self-care, sleep, diet, and unprotected sexual activity that occur in relation to substance use [19,24,25,28] may worsen disease. Pain and mental health comorbidities are prevalent among YCMC[29] and may heighten risk for substance use, including by adolescents who are seeking symptom relief.[27,30] Frequent healthcare visits, scheduling disruptions for procedures, and periods of feeling unwell due to flares and treatment side effects can interfere with participation in school and activities, contributing to disparities in educational attainment observed among YCMC in nationally representative longitudinal studies.[31]

In the U.S., insight about the negative educational impacts of substance use draws largely on national survey data that quantifies substance-specific risks, which limits understanding about the relative impacts of specific substances. Routine healthcare screening is tuned to determine health rather than educational impacts.[17,32] Yet understanding associations among use of specific substances and educational impacts is vital for accurate and specific health messaging and guidance. Norms and attitudes toward substances are changing, as are patterns of promotion and availability.[33,34] Discernment of life course relevant risks from substance use, including those related to educational impacts that presage future health and development, will inform efforts to ameliorate the U.S. national crisis in adolescent mental and behavioral health.[35]

This study sought to understand associations between use of alcohol, cannabis, and nicotine and indicators of negative educational impact relevant to adolescent well-being, testing the hypothesis of difference in associations for these substances. Additionally, this study sought to investigate differential associations in the relationship between educational impacts and substance use for youth with and without chronic medical conditions, testing the hypothesis of differential negative effects for YCMC.

METHODS

Study Sample

Between February 3, 2016, and November 20, 2018, via convenience sampling, adolescents aged 14−18 years who were in care at a general adolescent or subspecialty clinic for type 1 diabetes (endocrinology), juvenile idiopathic arthritis or systemic lupus erythematosus (rheumatology), or inflammatory bowel disease (including ulcerative colitis or Crohn's disease, gastroenterology) at a large teaching hospital in the Northeastern U.S. were recruited. Under the approval of the Boston Children's Hospital Institutional Review Board, adolescents ages 14−17 years provided consent to participate in the study with a waiver of parental permission, and those aged 18 years provided informed consent.

N=1,904 youth were recruited, of which n=975 (51.2%) assented/consented and provided complete data (n=958, 98.3%) on educational impacts (dependent variables) and past 12-month use of alcohol, cannabis, and nicotine (independent variables). The final analytic sample included n=445 (46.5%) participants without missing data from a specialty care clinic and n=513 (53.5%) from the adolescent medicine clinic (Figure 1). There was no sociodemographic difference between the youth included and those excluded in the analyses, except that the latter were less likely to self-report having 2-parent households (Appendix Table 1).

Measures

Self-reported survey data were collected using the Research Electronic Data Capture (REDCap) software platform.[36] Four distinct measures of educational impact were assessed (Appendix Table 2), including grades in the last report card (adapted from the Youth Risk Behavior Survey[37]), past 3-month truancy from classes or

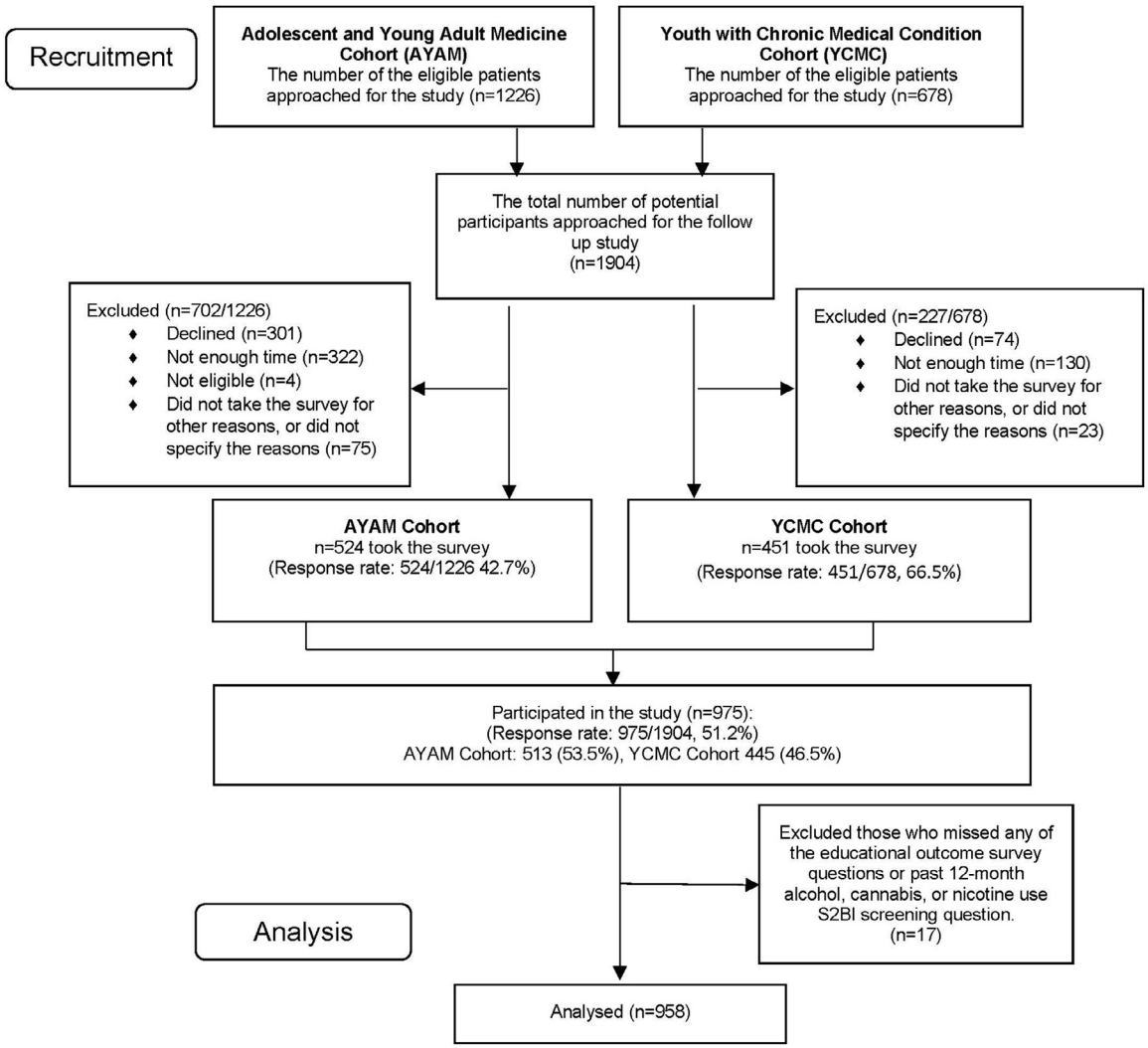

Figure 1. Flowchart for inclusion of the study participants. AYAM, Adolescent/Young Adult Medicine; S2BI, Screening to Brief Intervention; YCMC, Youth with Chronic Medical Conditions.

school and from extracurricular activities (adapted from the Healthy Youth Survey[38]), past 3-month detention (adapted from the National Longitudinal Survey of Youth[39]). Experience of any recent negative educational impact was defined as reporting any versus none of these four measures.

The Screening to Brief Intervention (S2BI)[40] tool was used to assess self-reported past 12-month use of alcohol, cannabis, nicotine, illegal drugs, prescription medication, and herbs or synthetic drugs.

The frequency of days used over the past three months was assessed for youth who reported past 12-month use of alcohol and cannabis.[41] Past 30-day use of alcohol and cannabis was assessed using a binary variable (any/none). Heavy episodic drinking was determined using age/gender-specific cutoffs.[42,43] Past 12-month substance use was categorized as "never" or "any (once or twice/monthly/weekly)" and these categories were used as predictors in the analyses.

Participants reported age, gender (dichotomized as female vs. male/other for regression analyses), race and ethnicity, household composition, and the highest level of education attained by a parent. Participants were also identified by clinic (dichotomized as "adolescent medicine" or "subspecialty clinic" for analyses). Participants in adolescent medicine were asked whether they have any current medical conditions (for sensitivity analyses, Appendix Table 3), and were also assessed for symptoms of depression and anxiety using the Patient Health Questionnaire-2 (PHQ-2)[44] and Generalized Anxiety Disorder-2 (GAD-2),[45] and queried self-rated health.[46]

Statistical Analyses

A complete case analysis approach was adopted for reports about educational impact measures and past 12-month alcohol, cannabis, and nicotine use. Summary statistics were generated using Pearson's Chi-squared test, Wilcoxon rank sum test, and Fisher's exact test (Table 1). Separate multivariate modified Poisson regression analyses[47−49] with robust error variance were used to examine associations among the past 12-month substance use measures and experience of any and each specific recent educational impact measure (Table 2, Appendix Table 4). Similarly, the association between past 30-day use of alcohol and cannabis and educational impact measures was assessed (Table 2). Unadjusted models were implemented separately for past 12-month alcohol, cannabis, or nicotine use (Model 1), then these models were adjusted for sociodemographic and health factors (Model 2), and measures of other substance use (Models 3-5) such that the final adjusted models included past 12-month alcohol, cannabis, nicotine, and the covariates (Appendix Table 5). As a sensitivity analysis, the lower grades outcome was assessed with a different cut-off (Appendix Table 6). Differential associations between past 12-month substance use and negative educational impacts for YCMC versus other youth were tested using an interaction term (Table 3).

To better understand whether associations among substance use and negative educational impacts were greater for more frequent or heavier substance use, sensitivity analyses were conducted using: an ordinal scale (S2BI) among all, reports of past 3-month use frequencies among youth with past year use by substance (Appendix Tables 7−9), and by examining associations for negative educational impacts measured using an aggregate ordinal scale (Appendix Table 10). Finally, associations for participants who reported any versus no chronic medical condition were examined to account for chronic illness experience among youth from adolescent medicine (Appendix Table 11). The statistical analyses used SAS 9.4 for interaction analyses, R (version 4.3.1), and R Studio (version 1.4.1717) for data analysis, using a statistical significance threshold of $p<0.05$.

RESULTS

N=958 adolescents were analyzed (mean age 16.0 years, SD 1.3, 564 (58.9%) female gender, and 481 (50.2%) White non-Hispanics) (Table 1). All participants from subspecialty clinics (i.e., YCMC) had either type 1 diabetes, a rheumatologic or inflammatory bowel disease, while 169 (32.9%) of general adolescent medicine clinic participants (n=513) reported at least one current medical condition (e.g., asthma). Overall, 294 (30.7%), 220 (23%), and 126 (13.2%) youth reported past 12-month use of alcohol, cannabis, and nicotine (e-cigarette or tobacco), respectively. Overall, 210 (21.9%) reported using 2 or more substances in the past 12-months, and 164 (17.1%) reported using both alcohol and cannabis in the past 12-months. Among youth who reported past 12-month alcohol use (n=294), 232 (78.9%) reported past 3-month alcohol use, and among those who reported past 12-month cannabis use (n=220), 177 (80.5%) reported past 3-month cannabis use.

The most common negative educational impact was receiving lower grades in school (i.e., mostly C's/D's/F's) reported by 210 (21.9%), followed by past 3-month truancy from extracurricular activities reported by 170 (17.7%), or school 164 (17.1%), and past 3-month detention reported by 82 (8.6%). More than two-fifths (407 (42.5%)) of the sample reported at least one negative educational impact (Table 1), and on average, compared to youth who did not report any, youth in this group were more likely to report older age (mean 16.2, SD 1.3 vs. 15.9, SD 1.4, P=0.005), being from a racially or ethnically minoritized group (68.3% vs. 36.1%, $p<0.001$), from a single/foster parent household (45.7% vs. 21.8%, $p<0.001$), less parental college education (51.4% vs. 27.2%, $p<0.001$), having symptoms of depression (16.2% vs. 6.4%, $p<0.001$) or anxiety 22.4% vs. 69 (12.5%, $p<0.001$), and in fair/poor health (19.4% vs. 11.3%, $p<0.001$). In bivariate analyses, past 12-month use of alcohol, cannabis, and nicotine each had associations with experiencing any negative educational impact (Table 1). In models that fully adjusted for sociodemographic factors and simultaneously included measures of past 12-month alcohol, cannabis, and nicotine use, past 12-month use of cannabis was associated with experience of three specific negative educational impact measures; adjusted prevalence ratios [APR] recent lower grades (Mostly C's/D's/F's) [APR] 1.54 [95% CI, 1.13−2.11], past 3-month truancy from school [APR] 2.16 [95% CI, 1.52−3.07], or detention [APR] 2.29 [95% CI, 1.33−3.94], and any negative educational impact [APR] 1.22 [95% CI, 1.02−1.46] (Table 2, Appendix Tables 4 and 5). In adjusted models, neither alcohol nor nicotine use was associated with any specific negative educational impact, while nicotine use was associated with experiencing any negative educational impacts [APR] 1.30 [95% CI, 1.07−1.57]. Past 12-month cannabis use maintained significant associations even after adjusting for past 12-month use of alcohol, nicotine, or other substances (Table 2, Appendix Tables 4 and 5). The findings were similar for past 30-day alcohol use and cannabis use, respectively (Table 2). In sensitivity analyses that employed different cut-offs for lower grades, only past 12-month cannabis use was associated with reporting

Table 1. Sample Characteristics by Negative Educational Impacts (n=958)[a]

Characteristic	Overall, n=958	Any negative educational impact[b], n (%) Any, n=407	Recent grades[c], n (%) mostly C's/D's/F's, n=210	Past 3-month truancy from activities[d], n (%) Any, n=170	Past 3-month truancy from school[e], n (%) Any, n=164	Past 3-month detention[f], n (%) Any, n=82
Clinic						
Adolescent general medicine	**513 (53.5%)**	**286 (70.3%)**	**153 (72.9%)**	**135 (79.4%)**	**113 (68.9%)**	**65 (79.3%)**
Subspecialty clinics	**445 (46.5%)**	**121 (29.7%)**	**57 (27.1%)**	**35 (20.6%)**	**51 (31.1%)**	**17 (20.7%)**
Current medical condition, any	**614 (64.1%)**	**210 (51.6%)**	**104 (49.5%)**	**79 (46.5%)**	**87 (53.0%)**	**40 (48.8%)**
Age, mean, (SD)	**16.0 (1.3)**	**16.2 (1.3)**	16.1 (1.3)	**16.2 (1.3)**	**16.6 (1.2)**	**15.6 (1.3)**
Gender						
Male	382 (39.9%)	177 (43.5%)	**105 (50.0%)**	59 (34.7%)	59 (36.0%)	39 (47.6%)
Female	564 (58.9%)	224 (55.0%)	**101 (48.1%)**	108 (63.5%)	103 (62.8%)	41 (50.0%)
Other	12 (1.3%)	6 (1.5%)	**4 (1.9%)**	3 (1.8%)	2 (1.2%)	2 (2.4%)
Race/ethnicity						
White non-Hispanic	**481 (50.2%)**	**129 (31.7%)**	**60 (28.6%)**	**44 (25.9%)**	**49 (29.9%)**	**21 (25.6%)**
Other	**477 (49.8%)**	**278 (68.3%)**	**150 (71.4%)**	**126 (74.1%)**	**115 (70.1%)**	**61 (74.4%)**
Parent college degree						
Less than a college degree	**359 (37.5%)**	**209 (51.4%)**	**132 (62.9%)**	**87 (51.2%)**	**83 (50.6%)**	**47 (57.3%)**
Household composition						
Single parent/no parent/foster care	**306 (31.9%)**	**186 (45.7%)**	**110 (52.4%)**	**77 (45.3%)**	**83 (50.6%)**	**42 (51.2%)**
Depression screening, positive (PHQ-2 ≥3)	**101 (10.5%)**	**66 (16.2%)**	**40 (19.0%)**	**30 (17.6%)**	**35 (21.3%)**	**14 (17.1%)**
General anxiety screening, positive (GAD-2 ≥3)	**160 (16.7%)**	**91 (22.4%)**	**51 (24.3%)**	**40 (23.5%)**	**42 (25.6%)**	16 (19.5%)
Self-rated health						
Fair/poor	**141 (14.7%)**	**79 (19.4%)**	**52 (24.8%)**	26 (15.3%)	**39 (23.8%)**	11 (13.4%)
Excellent/very good/good	**817 (85.3%)**	**328 (80.6%)**	**158 (75.2%)**	144 (84.7%)	**125 (76.2%)**	71 (86.6%)
Past 12-month alcohol use, any	**294 (30.7%)**	**160 (39.3%)**	75 (35.7%)	**75 (44.1%)**	**85 (51.8%)**	31 (37.8%)
Past 3-month alcohol use frequency, mean (SD)	**0.92 (2.63)**	**1.15 (3.01)**	**0.91 (2.87)**	**1.26 (3.07)**	**1.73 (3.97)**	0.95 (2.00)
Past 3-month alcohol use, any	**232 (24.2%)**	**124 (30.5%)**	56 (26.7%)	**55 (32.4%)**	**63 (38.4%)**	26 (31.7%)
Past 30-day alcohol use, any	**156 (16.3%)**	**84 (20.6%)**	41 (19.5%)	34 (20.0%)	**46 (28.0%)**	18 (22.0%)
Past 12-month cannabis use, any	**220 (23.0%)**	**141 (34.6%)**	**77 (36.7%)**	**58 (34.1%)**	**87 (53.0%)**	**33 (40.2%)**
Past 3-month cannabis use frequency, mean (SD)	**2.8 (11.2)**	**4.6 (13.9)**	**5.8 (15.6)**	**3.1 (10.2)**	**7.9 (17.3)**	**4.3 (11.3)**
Past 3-month cannabis use, any	**177 (18.5%)**	**115 (28.3%)**	**66 (31.4%)**	**48 (28.2%)**	**71 (43.3%)**	**29 (35.4%)**
Past 30-day cannabis use, any	**137 (14.3%)**	**93 (22.9%)**	**57 (27.1%)**	**34 (20.0%)**	**60 (36.6%)**	**23 (28.0%)**
Past 12-month nicotine use, any[g]	**126 (13.2%)**	**76 (18.7%)**	35 (16.7%)	**32 (18.8%)**	**42 (25.6%)**	13 (15.9%)
Past 12-month other substance use, any[h]	**51 (5.3%)**	**35 (8.6%)**	**19 (9.0%)**	**15 (8.8%)**	**18 (11.0%)**	8 (9.8%)

Notes: Column percentages are shown. Boldface indicates statistical significance (p<0.05).

[a]p value testing for the difference between youth who reported negative educational impact and those who did not, using Pearson's chi-squared test, Wilcoxon rank sum test, and Fisher's exact test.

[b]Having experienced any of grades, mostly C's/D's/F's in the last report card, past 3-month detention, truancy from activities/school during the past 3 months.

[c]What were your grades on your last report card in school? The response ranged from "Mostly A's and/or B's" to "Mostly D's and/or F's," and the response was dichotomized as "Mostly C's/D's/F's" vs. "Mostly A's/B's."

[d]In the past 3 months, how often did you skip extracurricular activities (activities done after school or on the weekend such as sports, clubs, etc.) without a parent/guardian's knowledge? The response ranged from Never/once or twice/monthly/weekly and the response was dichotomized as "any" vs. "never."

[e]In the past 3 months, how often did you skip classes or school without a parent or guardian's permission? The response ranged from Never/once or twice/monthly/weekly, and the response was dichotomized as "any" vs. "never."

[f]In the past 3 months, how many days were you given detention or in-school suspension? The response ranged from 0 to 10 days or more, and the response was dichotomized as "any" vs. "never."

[g]Past 12-month nicotine use was operationalized as either any e-cigarette or tobacco use during the past 12 months.

[h]Other substances, including prescription drugs (e.g., pain medication or Adderall), illegal drugs (such as cocaine, Ecstasy, Molly), or herbs or synthetic drug or substances (such as salvia, "K2", or bath salts).

GAD, General Anxiety Disorder; PHQ, Patient Health Questionnaire.

Table 2. Associations Between Substance Use and Negative Educational Impacts (n=958)

Outcome	Any negative educational impact: Any (vs. none[ref])[a]		Most recent grades in the last report card: Mostly C's/D's/F's (vs. mostly A's/B's [ref])[b]		Past 3-month truancy from activities: Any (vs. never[ref])[c]		Past 3-month truancy from school: Any (vs. never[ref])[d]		Past 3-month detention: Any (vs. never[ref])[e]	
Predictor	**APR (95% CI)**	***p*-value**	**APR (95% CI)**	***p*-value**	**APR (95% CI)**	***p*-value**	**APR (95% CI)**	***p*-value**	**APR (95% CI)**	***p*-value**
Past 12-month alcohol use[f]										
Any	1.15 (0.96, 1.38)	0.13	0.99 (0.73, 1.36)	0.97	1.38 (0.98, 1.96)	0.069	1.22 (0.87, 1.70)	0.25	1.16 (0.67, 2.01)	0.59
Never (ref.)	—		—		—		—		—	
Past 12-month cannabis use[g]										
Any	**1.22 (1.02, 1.46)**	**0.031**	**1.54 (1.13, 2.11)**	**0.007**	0.99 (0.70, 1.41)	0.97	**2.16 (1.52, 3.07)**	**<0.001**	**2.29 (1.33, 3.94)**	**0.003**
Never (ref.)	—		—		—		—		—	
Past 12-month nicotine use[h]										
Any	**1.3 (1.07, 1.57)**	**0.007**	1.08 (0.75, 1.57)	0.68	1.38 (0.94, 2.01)	0.1	1.3 (0.94, 1.81)	0.11	0.91 (0.49, 1.67)	0.75
Never (ref.)	—		—		—		—		—	
Past 30-day alcohol use[i]										
Any	1.16 (0.97, 1.39)	0.095	1.22 (0.89, 1.67)	0.22	1.02 (0.71, 1.48)	0.9	1.26 (0.90, 1.75)	0.18	1.37 (0.76, 2.45)	0.29
Never	—		—		—		—		—	
Past 30-day cannabis use[j]										
Any	**1.25 (1.06, 1.47)**	**0.008**	**1.75 (1.32, 2.32)**	**<0.001**	1 (0.70, 1.43)	>0.99	**1.91 (1.39, 2.63)**	**<0.001**	**2.21 (1.24, 3.94)**	**0.007**
Never	—		—		—		—		—	

Notes: Boldface indicates statistical significance (p<0.05)

[a]Having experienced any of detention, skipping activities/school during the past 3months, or grades mostly C's/D's/F's in the last report card.

[b]Survey question: "What were your grades on your last report card in school?" The response ranged from "Mostly A's and/or B's" to "Mostly D's and/or F's" and the response was dichotomized as "Mostly C's/D's/F's" vs. "Mostly A's/B's".

[c]Survey question: "In the past 3 months, how often did you skip extracurricular activities (activities done after school or on the weekend such as sports, clubs, etc.) without a parent/guardian's knowledge?" The response ranged from Never/once or twice/monthly/weekly and the response was dichotomized as "any" vs. "never".

[d]Survey question: "In the past 3 months how often did you skip classes or school without a parent or guardian's permission?" The response ranged from Never/once or twice/monthly/weekly and the response was dichotomized as "any" vs. "never".

[e]Survey question: "In the past 3 months, how many days were you given detention or in-school suspension? The response ranged from 0 to 10 days or more, and the response was dichotomized as "any" vs. "never".

[f]The fully adjusted models included the past 12-month alcohol use, adjusting for past 12-month cannabis use, nicotine use, other substance use, and sociodemographics including age, clinic types, gender (dichotomized as females vs. other), race and ethnicity, parental education, household composition, depression and anxiety screening results, and self-rated health.

[g]The fully adjusted models included the past 12-month cannabis use, adjusting for past 12-month alcohol use, nicotine use, other substance use, and sociodemographics including age, clinic types, gender (dichotomized as females vs. other), race and ethnicity, parental education, household composition, depression and anxiety screening results, and self-rated health.

[h]The fully adjusted models included the past 12-month nicotine use, adjusting for past 12-month alcohol use, cannabis use, other substance use, and sociodemographics including age, clinic types, gender (dichotomized as females vs. other), race and ethnicity, parental education, household composition, depression and anxiety screening results, and self-rated health.

[i]The fully adjusted models included the past 30-day alcohol use, adjusting for past 30-day cannabis use, past 12-month nicotine use, other substance use, and sociodemographics including age, clinic types, gender (dichotomized as females vs. other), race and ethnicity, parental education, household composition, depression and anxiety screening results, and self-rated health.

[j]The fully adjusted models included the past 30-day cannabis use, adjusting for past 30-day alcohol use, past 12-month nicotine use, other substance use, and sociodemographics including age, clinic types, gender (dichotomized as females vs. other), race and ethnicity, parental education, household composition, depression and anxiety screening results, and self-rated health.

APR, adjusted prevalence ratio.

Table 3. Associations Between Substance Use and Negative Educational Impacts Stratified by Chronic Condition[a] (*n*=958)

Outcome	General adolescent medicine clinic participants		Subspecialty clinic participants (with a chronic condition)		*p*-value for interaction
	APR (95% CI)	*p*-value	APR (95% CI)	*p*-value	
Predictor: Past 12-month alcohol use[b,c]					
Any negative educational impact: any[d]	1.06 (0.88, 1.28)	0.53	**1.45 (1.05, 1.98)**	**0.023**	0.06
Most recent grades in the last report card: mostly C's/D's/F's[e]	0.96 (0.69, 1.35)	0.82	1.12 (0.65, 1.91)	0.68	0.60
Past 3-month truancy from activities: any[f]	1.25 (0.87, 1.8)	0.22	**2.15 (1.1, 4.21)**	**0.025**	0.12
Past 3-month truancy from school: any[g]	1.04 (0.73, 1.49)	0.83	**1.82 (1.06, 3.12)**	**0.03**	0.06
Past 3-month detention: any[h]	1.09 (0.64, 1.86)	0.75	1.63 (0.50-5.33)	0.42	0.49
Predictor:Past 12-month cannabis use[c,i]					
Any negative educational impact: any[d]	1.06 (0.88, 1.27)	0.56	**1.96 (1.45, 2.65)**	**<0.0001**	**<0.0001**
Most recent grades in the last report card: mostly C's/D's/F's[e]	**1.50 (1.07, 2.09)**	**0.018**	**1.75 (1.03, 2.95)**	**0.038**	0.58
Past 3-month truancy from activities: any[f]	0.91 (0.63, 1.32)	0.61	1.62 (0.82, 3.19)	0.16	0.11
Past 3-month truancy from school: any[g]	**1.79 (1.23, 2.58)**	**0.002**	**3.53 (2.01, 6.19)**	**<0.0001**	**0.020**
Past 3-month detention: any[h]	**1.90 (1.12, 3.20)**	**0.017**	**6.33 (2.14, 18.7)**	**0.001**	**0.026**
Predictor:Past 12-month nicotine use[c,j]					
Any negative educational impact: any[d]	1.05 (0.85, 1.3)	0.64	**1.88 (1.39, 2.54)**	**<0.0001**	**0.001**
Most recent grades in the last report card: mostly C's/D's/F's[e]	0.86 (0.54, 1.36)	0.52	1.65 (0.99, 275)	0.05	**0.041**
Past 3-month truancy from activities: any[f]	1.16 (0.75, 1.78)	0.50	**2.26 (1.16, 4.42)**	**0.016**	0.08
Past 3-month truancy from school: any[g]	1.12 (0.76, 1.66)	0.56	**1.72 (1.02, 2.91)**	**0.042**	0.17
Past 3-month detention: any[h]	0.99 (0.51, 1.92)	0.99	0.60 (0.14, 2.50)	0.49	0.52
Predictor: Past 30-day alcohol use[k,l]					
Any negative educational impact: any[d]	1.10 (0.91, 1.33)	0.33	**1.39 (0.99, 1.95)**	0.06	0.20
Most recent grades in the last report card: mostly C's/D's/F's[e]	1.19 (0.85, 1.67)	0.31	1.33 (0.72, 2.44)	0.36	0.73
Past 3-month truancy from activities: any[f]	0.83 (0.55, 1.26)	0.37	**2.34 (1.18, 4.63)**	**0.015**	**0.006**
Past 3-month truancy from school: any[g]	1.09 (0.75, 1.58)	0.66	**1.78 (1.01, 3.14)**	**0.046**	0.12
Past 3-month detention: any[h]	1.34 (0.74, 2.41)	0.34	1.58 (0.42, 5.91)	0.49	0.80

(*continued on next page*)

Table 3. Associations Between Substance Use and Negative Educational Impacts Stratified by Chronic Condition[a] (*n*=958) (*continued*)

Outcome	General adolescent medicine clinic participants		Subspecialty clinic participants (with a chronic condition)		*p*-value for interaction
	APR (95% CI)	*p*-value	APR (95% CI)	*p*-value	
Predictor:Past 30-day cannabis use[l,m]					
Any negative educational impact: any[d]	1.11 (0.94, 1.32)	0.22	**1.88 (1.38, 2.58)**	**<0.0001**	**0.001**
Most recent grades in the last report card: mostly C's/D's/F's[e]	**1.73 (1.30, 2.32)**	**0.0002**	**1.83 (1.04, 3.22)**	**0.036**	0.85
Past 3-month truancy from activities: any[f]	0.84 (0.56, 1.26)	0.39	**2.32 (1.15, 4.68)**	**0.019**	**0.008**
Past 3-month truancy from school: any[g]	**1.76 (1.24, 2.50)**	**0.002**	**2.51 (1.42, 4.41)**	**0.002**	**0.24**
Past 3-month detention: any[h]	**1.82 (1.02, 3.26)**	**0.043**	**7.70 (2.32, 25.54)**	**0.001**	**0.014**

Notes: Boldface indicates statistical significance ($p<0.05$)

[a]Subspecialty clinic patients are under the care for type 1 diabetes (endocrinology), juvenile idiopathic arthritis or systemic lupus erythematosus (rheumatology), or inflammatory bowel disease (including ulcerative colitis or Crohn's disease, gastroenterology).

[b]The fully adjusted models included the interaction term with chronic condition status and past 12-month alcohol use, adjusting for past 12-month cannabis use, nicotine use, other substance use, and sociodemographics including age, gender, race and ethnicity, parental education, household composition, depression and anxiety screening results, and self-rated health.

[c]The measure assesses whether the association between the past 12-month substance use and negative educational impacts is uniform across different levels of chronic condition status (between general adolescent medicine and subspecialty clinics). The P-value is for the difference by chronic condition status (e.g., if $p<0.05$, the association between the past 12-month substance use and the negative educational impact differs depending on the chronic condition status).

[d]Having experienced any of detention, skipping activities/school during the past 3 months, or grades mostly C's/D's/F's in the last report card. The reference is "none".

[e]Survey question: "What were your grades on your last report card in school?" The response ranged from "Mostly A's and/or B's" to "Mostly D's and/or F's" and the response was dichotomized as "Mostly C's/D's/F's" vs. "Mostly A's/B's." The reference is "mostly A's/B's"

[f]Survey question: "In the past 3 months, how often did you skip extracurricular activities (activities done after school or on the weekend such as sports, clubs, etc.) without a parent/guardian's knowledge?" The response ranged from Never/once or twice/monthly/weekly and the response was dichotomized as "any" vs. "never." The reference is "never".

[g]Survey question: "In the past 3 months how often did you skip classes or school without a parent or guardian's permission?" The response ranged from Never/once or twice/monthly/weekly and the response was dichotomized as "any" vs. "never."

[h]Survey question: "In the past 3 months, how many days were you given detention or in-school suspension? The response ranged from 0 to 10 days or more, and the response was dichotomized as "any" vs. "never."

[i]The fully adjusted models included the interaction term with chronic condition status and past 12-month cannabis use, adjusting for past 12-month alcohol use, nicotine use, other substance use, and sociodemographics including age, gender, race and ethnicity, parental education, household composition, depression and anxiety screening results, and self-rated health.

[j]The fully adjusted models included the interaction term with chronic condition status and nicotine use, adjusting for past 12-month alcohol use, cannabis use, other substance use, and sociodemographics including age, gender, race and ethnicity, parental education, household composition, depression and anxiety screening results, and self-rated health.

[k]The fully adjusted models included the interaction term with chronic condition status and past 30-day alcohol use, adjusting for 30-day cannabis use, past 12-month nicotine use, other substance use, and sociodemographics including age, gender, race and ethnicity, parental education, household composition, depression and anxiety screening results, and self-rated health.

[l]The measure assesses whether the association between the past 30-day substance use and negative educational impacts differs across different levels of chronic condition status (between general adolescent medicine and subspecialty clinics). The P-value is for the difference by chronic condition status (e.g., if $p<0.05$, the association between the past 30-day substance use and the negative educational impact differs depending on the chronic condition status).

[m]The fully adjusted models included the interaction term with chronic condition status and past 30-day cannabis use, adjusting for past 30-day alcohol use, past 12-month nicotine use, other substance use, and sociodemographics including age, gender, race and ethnicity, parental education, household composition, depression and anxiety screening results, and self-rated health.

APR, adjusted prevalence rate ratio.

Mostly D's/F's: [APR], 3.23; [95% CI, 1.64−6.36] (Appendix Table 6).

YCMC were more likely than their peers without a chronic condition to report any negative educational impact ($p<0.001$), past 3-month truancy from school ($p=0.020$), and past 3-month detention ($p=0.026$), in association with past 12-month cannabis use (Table 3). The results were similar in analyses using past 30-day alcohol and cannabis use as predictors (Table 3). YCMC were also more likely to report any negative educational impact and poor grades in association with past 12-month nicotine use, respectively ($p<0.001$, $p=0.041$).

In sensitivity analyses, the level of past-year cannabis use was positively associated with all but one impact measure (Appendix Table 7), consistent with main analyses. Among youth with past-year use, an increase of one day of past 3-month use of alcohol or cannabis was not associated with increased prevalence rates of negative educational impacts (Appendix Tables 8 and 9).

Consistent with main outcome analyses (Table 3), there was a significant interaction between cannabis use and chronic medical condition status on negative educational impacts, including any negative educational impact, school truancy, and detention in analyses of participants with any chronic medical condition (i.e., from adolescent medicine or specialty clinics) versus participants reporting none (Appendix Table 10). Sensitivity analyses assessing the association between past 12-month substance use and the number of reported negative educational impacts showed that cannabis use specifically was associated with the increase in the number of negative educational impact outcomes (Appendix Table 11).

DISCUSSION

In a large, medically heterogeneous sample of adolescents, substance use was associated with experience of several specific negative educational impacts. In analyses that estimated associations in relation to past 12-month use of alcohol, cannabis, and nicotine, cannabis use appeared to be the driver of observed associations. These findings are consistent with reports of the detrimental effects of cannabis use on cognitive performance and educational measures,[10,20,21] an area of active research and inconsistent results.[6] Results also align with those from longitudinal studies that have found negative effects of cannabis use on participation in school and other activities among males.[50] Findings are important given increasingly prevalent views that cannabis use carries few risks,[51] is helpful for addressing stress and anxiety and can even cure medical conditions such as diabetes − a false belief.[52] Societal tendencies toward discounting of potential harms and assertion of benefit combined with legalization and commercialization have contributed to sharp declines in levels of perceived riskiness that otherwise protect youth from cannabis use.[53] As such, understanding youth experience of cannabis-related harms including in sectors, such as education that are not typically included in clinical evaluation or public health surveillance, is especially important. Interestingly, among youth reporting past year cannabis use, associations were not found between increased prevalence of negative educational impacts and increased frequency of use, which may reflect limited variability or statistical power. Nevertheless, findings suggest the importance of public health messaging and policy controls targeting no or reduced cannabis use by adolescents.

The educational impacts chosen for this project have many inputs and are inter-related. Cannabis use impacts cognition and effects have been observed to last up to 6 weeks among cannabis-dependent youth in treatment with supervised abstention.[54] As such, regular cannabis use may have continuous impacts that are difficult to notice but nonetheless influence performance on exams and school assignments. Cannabis use has long been associated with amotivation,[55] which may lead to decreased school participation, with attendant disciplinary consequences and lower grades. Near daily cannabis use including during the school day is reported by 3.1 percent of high school students,[2] and being impaired during the school day or possessing drugs or paraphernalia may increase the likelihood of disciplinary consequences and also impact academic performance.

Globally, cannabis is the most prevalent illicit substance used by adolescents,[56] and the most common reason for admission to a substance use disorder treatment program among youth.[57] In the U.S., state and local policies governing cannabis sale and consumption have become more permissive, and promotion of cannabis as safe and health benefiting is common.[58] Compared to a decade ago, cannabis has become easier to access, more available and more potent.[59] Among U.S. high school youth, the overall prevalence of cannabis use increased markedly though intermittently in the decade preceding 2021.[2] Rising levels of use were preceded by large declines in the percentage of youth reporting they perceive cannabis use as risky or dangerous,[2,53] and by rapid changes in social factors affecting the use of cannabis.

A disproportionately high risk for reporting negative educational impacts in relation to cannabis use was identified among youth recruited with a chronic medical condition. The study findings support the importance of implementing substance use prevention and mitigation

efforts for YCMC, a group at uniquely high risk of harms.[26,60] National longitudinal studies show that YCMC are at disproportionately high risk of experiencing negative educational outcomes as they grow up, despite holding equivalent aspirations and expectations for educational attainment as healthy youth in childhood.[31] Other studies have shown that many YCMC report using cannabis for symptom relief and these youth exhibit substance use patterns associated with risk for substance use disorder.[30] Together with current findings, results suggest that cannabis use poses uniquely high risks for YCMC, a group whose educational attainment may already be jeopardized by interruptions stemming from disease and treatment burdens and substance use patterns that tend toward greater severity.[24]

In sum, findings add to the growing body of work characterizing the negative consequences for youth associated with cannabis use. Information is relevant to clinicians in healthcare and school settings who may incorporate information about cannabis related harms into guidance provided to patients and families, as recommended by the American Academy of Pediatrics,[61] and for researchers aiming to understand the practical impacts of interventions intended to reduce cannabis use.

Limitations

This report reflects a cross-sectional investigation of associations that cannot determine causation. Longitudinal studies are needed to better ascertain the temporal unfolding of poor educational outcomes. Study information relied on self-report, which is standard for survey studies but subject to recall bias. Future studies could catalogue outcomes by drawing from ecological data (i.e., school records and attendance/participation logs).[62] The study is based on a convenience sample that is relatively large and from multiple clinics with similar substance use rates to those found in denominator-based studies. However, participants opted-in to this study and are from a single institution, limiting generalizability. Some meaningful educational impacts, including longitudinal measures of achievement and school completion/drop-out were not assessed. Although the study included relevant covariates, there may be unmeasured confounding, including from family, neighborhood, and peer factors. Finally, surveys do not include information on physiological impacts of substances on neurological functioning.

CONCLUSIONS

In this study, cannabis use was associated with a heightened risk for self-report of negative educational impacts, even after controlling for use of alcohol and nicotine. As cannabis becomes more acceptable and available, greater investment is needed in tracking harms related to use across social sectors and subgroups. Accurate information about cannabis related risks may help inform health policies, decision-making, and prevention programs to improve adolescent wellbeing.

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SUPPLEMENTAL MATERIAL

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